

Software Developer

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🌐 <https://kdnbradshaw.click>

🐙 Github:

EDUCATION

BRIGHAM YOUNG UNIVERSITY:

- Computer Science:
Bioinformatics
Emphasis (BS)
- Dec. 2022 - Apr 2027
- BYU Developers Club

Engineering Profile

Languages: Python, Java,
C/C++, SQL, R, Javascript,
React, Node.js

Software Engineering: Data
Structures and Algorithms,
Object-oriented design, API
development, Git, Machine
Learning

Testing & Reliability:
Test-driven development,
automated testing, PyTest

Data & Scientific Computing:
Statistical analysis, linear
regression, tidyverse/dplyr,
structured data processing

Web & Tooling: REST-style
services, WebSockets,
backend systems,
React/JavaScript, HTML/CSS

Kaden Bradshaw

APPLIED EXPERIENCE:

Distributed Client-Server Game System - Java, SQL, WebSockets

Architected a real client-server system (1,000+ lines of code) simulating a real-time multiplayer environment
Designed and implemented 10+ RESTful API endpoints with SQL-backed storage for user sessions, game state, and move history
Built a WebSocket communication layer to enable low-latency multi-client state synchronization and state broadcasting
Authored 40+ automated integration and unit tests, following test-driven development principles
Managed version control, structured codebase, and technical documentation with Git

Python Utilities & Data Engineering - Python, R

Performed statistical analysis on 20+ biological databases using R to extract patterns and validate data integrity
Designed an image-processing pipeline for structured transformation from visual data
Engineered a scalable multi-page web crawler for data extraction and processing
Simulated a 2D gravity-based physics system modeling dynamic interactions
Optimized algorithms for correctness, and cross-module reliability in data-processing systems

BYU Developers Club - AI Knowledge Assistant - React, Node.js, NPM

Developed modular React UI components to improve interaction flow and front-end usability
Integrated external APIs to enable knowledge querying and response handling
Collaborated in a multi-developer environment using Git workflows, pull requests, and code reviews

PROFESSIONAL EXPERIENCE:

BYU Grounds Crew — Facilities & Systems Operations

Sept 2023 - Present

Performed routine inspection and upkeep of mechanical equipment
Followed structured operational procedures to ensure safety of employees

Greenix Pest Control — Operations & Team Coordination

Apr 2023- Aug 2023

Managed contracts and operational logistics with clients